Exploring the interaction between compact binaries and their host galaxy

Mark Janosdeak

Student ID: 10748961

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University of Manchester
Department of Physics and Astronomy

Abstract

This MPhys thesis investigates the behaviour of binary systems with one of the constituting components being a neutron star in their host galaxy. The galaxy's morphology was mapped using analytical functions best fitting the observed data of the Milky Way. An a priori guess was made for the star formation history of the Milky Way, assuming a non-changing relative mass distribution. From this, binary systems were created using random variate sampling techniques and their evolution was tracked, exploring orbital properties of the population.

1 Introduction

The multiplicity frequency in stellar astrophysics is the likelihood of being in a binary or a multiple star system. This metric has long intrigued astronomers and serves as a valuable tool from the theoretical standpoint. It has been determined that Population I stars with masses of > 8 $M_\odot$ are most likely to exist as a pair or a multiple star system (Duchêne & Kraus, 2013). The evolution of such systems can result in the formation of a compact binary system, defined as a binary system containing at least one compact object, such as a neutrons star (NS), black hole (BH) and a white dwarf (WD) (Tauris & van den Heuvel, 2006a). If the companion is a massive star, stellar matter can accrete onto the compact object. The transferred mass heats up as it descends the compact object's potential well, leading to the emission of high-energy X-rays (Burns Boyle, 1984).

Other types of compact binary systems include those where both companions are compact objects, such as a neutron star paired with a black hole or another neutron star, referred to as neutron star-neutron star (NSNS) and black hole-neutron star (BHNS). The orbit of such systems decays via the emission of gravitational waves until they merge. This, along with the ringdown, the emission of gravitational waves from the single BH to dissipate any distortions of its shape (Abadie et al., 2011), after a merger, is called a coalescence. When one of the components is a neutron star, such a coalescence can produce electromagnetic phenomena, namely short-duration gamma-ray bursts and a kilonova (Li & Paczyński, 1998). The latter is due to the radioactive decay of heavy elements formed during rapid neutron capture events (r-process). The former is less understood but believed to be the consequence of the production of relativistic jets during mergers (Kouveliotou et al., 1993).

The largest of these merger events have become detectable since the first ground-based gravitational wave detection (Abbott et al., 2016). Future space-based gravitational wave observatories, such as the Laser Interferometer Space Antenna (LISA), are expected to measure on the order of 10^7 compact binaries in our host galaxy (Kremer & Breivik, 2017). This provides a need for simulations of compact binary systems and an investigation into the possible trends surrounding them and their progenitors. This thesis project was inspired by the research carried out by Soheb Mandhai (Mandhai et al., 2022).

The primary limitation in investigating doubly compact binary systems stems from the current constraints in observing gravitational waves. Consequently, if a trend is predicted from simulations, it cannot be tested until a space-based gravitational wave detector is launched. However, an alternative type of compact binary system can be utilized, pulsars. We have identified over 2000 systems in the Milky Way galaxy, making them a more promising candidate for testing predictions. In this thesis I investigated the behaviour of binary systems that undergo a single natal kick. One of the main focuses of this work was to examine how these systems evolve over time when subjected to kicks from a range of magnitudes.

In this work, Section 2 covers the background theory of gravitational potentials, the evolution of binary stars and natal kicks. Section 3 outlines the implementation and modelling of the theory with a computer program. In Section 4, the structure and contents of the program are discussed, along with the obtained results. Section 5 addresses current assumptions and suggests further improvements. Finally, in Section 6, a summary of the project is given.

2 Theory

In this section the background theory is presented to elucidate the motivation and fundamental mathematics required to produce the desired computational model. It begins with a mathematical description of galactic potentials and progresses to a more qualitative description of the evolution of binary systems and natal kicks.

2.1 Building up galactic gravitational potentials

Inferring a mass distribution, and consequently, a gravitational potential distribution for a galaxy from observed data typically follows a specific procedure. Initially, an observed rotation curve for the galaxy is measured, and a best-fit curve is produced. Subsequently, numerous analytical expressions for gravitational potentials are iterated over their free parameters and integrated to produce a rotational curve. This curve is then compared to the experimentally determined one until the exact analytical expressions with their free parameters are established. A shift in the overall shape of the Milky Way's rotation curve (Jiao et al., 2023) has been observed since the Gaia Data Release 3 (Gaia Collaboration et al., 2023). Consequently, I chose to utilize the older conventional result as outlined in Bovy (2015).

That potential was formulated using three analytic profiles: the Navarro-Frenk White (NFW) profile for modelling the dark matter halo (Navarro et al., 1996), the Miyamoto-Nagai (MN) profile for the stellar matter and interstellar gas in the disk (Miyamoto & Nagai, 1975), and a spherical power-law (SPL) density with exponential cutoff for the bulge (Bovy & Rix, 2013).

The NFW potential profile is described by Equation 1, where G represents the gravitational constant, r is the spherical polar radius, ρ_0 and R_s are free parameters fixed by the process described above (with R_s being the scale radius). Together, G and ρ_0 establish an overall scale for the potential.

$$\Phi(r) = -\frac{4\pi G\rho_0 R_s^3}{r}\ln\left(1+\frac{r}{R_s}\right) \tag{1}$$

The MN potential is constructed according to Equation 2, where M is an overall scale factor, R and z are the radius and vertical distance in the cylindrical polar coordinate frame, and a and b are the scale length and the scale height respectively. The ration of b/a determines the flattening of the mass distribution, with $b/a \gg 1$ indicating an approximately spherical distribution and $b/a \ll 1$ describing a significantly flattened one.

$$\Phi(R, z) = -\frac{GM}{\sqrt{R^2 + (\sqrt{z^2+b^2}+a)^2}} \tag{2}$$

The SPL density with an exponential cutoff is defined in Equation 3, where α is the inner power, r_c is the cutoff radius and r_1 is the overall scale, these parameters are free to vary. On the left hand side of Equation 3, ρ represents a density distribution. To obtain the potential from it, Poisson's equation needs to be solved.

$$\rho(r) = \left(\frac{r_1}{r}\right)^{\alpha} \exp\left(-\left(\frac{r}{r_c}\right)^2\right) \tag{3}$$

2.2 The evolution of binary stars

In the past, double stars have been commonly mistaken for visual binaries. However, with the advancements of telescopes and observational techniques, the categorization of binary stars has expanded. The modern definition of a binary star is as follows: two stars that orbit a common center of mass, a concept first articulated by Sir William Herschel (Herschel, 1802), who also coined the term "binary star" in the same definition.

The prevailing theory regarding the formation of binary stars states that the vast majority of binary systems develop during star formation from molecular gas clouds. The observation of a binary protostar (Tobin et al., 2013) not yet on the main sequence, supports this hypothesis. While formation by a gravitational capture is not impossible, it is a highly unlikely scenario, given that around half the stars in our galaxy have been identified as a multiple star system (Abt & Levy, 1976).

Following the star formation period, a binary system most likely consists of two main sequence stars. Depending on their respective masses, the evolution of the system can take various directions. Examining the multiplicity frequency of heavy stars, the importance of binary evolution becomes evident in the overall stellar evolution (Tauris & van den Heuvel, 2006a). This significance is further emphasised by the finding that the majority of heavy mass stars found in a binary system will at some point during their evolution interact with each other (Sana et al., 2012), introducing new mechanisms to stellar evolution that do not affect single stars.

The Roche model was developed to describe the equilibrium surfaces of the gravitational potential between two point masses, distorted due to tidal forces and rotations. Additionally, the Roche lobe represents the outermost closed equipotential surface around each stellar component (Kopal, 1955). This surface allows the classification of binary systems based on the component sizes and distances from each other, resulting in three categories: detached, semidetached and contact binary stars.

Detached binary stars are systems where each star is within their corresponding Roche lobe, allowing the two stars to evolve separately. Most binary systems fall into this category (Loukaidou et al., 2021). Semidetached binaries are systems where only one of the components fills its Roche lobe (donor). Material from the donor is transferred to the other member of the pair (accretor) in a process called mass transfer, influencing their evolution. Further details about this process are discussed below. Finally, contact binaries are systems where both stars fill their Roche lobes, and the uppermost part of the stellar atmosphere forms a common envelope. This slows the orbital motion due to friction and eventually can lead to the merging of the component stars (Quintana & Lissauer, 2007).

2.3 Natal kicks

The stellar evolution of a massive star is a well-known process, culminating in a supernova explosion that produces a NS or a BH depending on the progenitor star's mass. Ultimately, nuclear fusion reactions cease, and after some intermediate steps, a compact object forms. For stars with masses ranging from the smallest to the mid-range, approximately $> 10M_{\odot}$ a WD remains. Progenitors heavier than this result in a supernova explosion, leaving behind a NS or a BH.

During the Type II supernovae, which primarily leaves behind a NS, a phenomenon called natal kick (pulsar kick) can occur. While there is no widely accepted explanation, some strong candidate theories exist. Natal kicks are defined as the velocity gained by the NS compared to the local standard of rest of the center of mass of its progenitor system or star, depending on the multiplicity. For isolated pulsars and NS, the magnitude of the kick is usually found to be between 200-500 kmh^{-1} (Igoshev & Verbunt, 2018; Igoshev, 2020). Observations suggest a bimodal distribution of natal kicks. One significant piece of evidence supporting this claim is the neutron star retention problem in globular clusters. Specifically, if all neutron stars produce a natal kick of the above mentioned magnitude, only around 1% of the NS would remain in globular clusters, where their progenitors were born. However, observations suggest a large number of pulsars located in globular clusters at the present age.

Furthermore, without considering significant mass transfer, applying the isolated NS natal kick to a one of the companions of binary systems results in the system's disruption (Pfahl et al., 2002). Mass transfer plays a crucial role in the evolution of binary systems, especially when at least one of the components is a compact object. The mechanism of mass transfer in a binary system is as follows: the heavier of the two main sequence stars burns through their fuel faster, and as it reaches the next stage of its evolution, it expands. If it surpasses its Roche lobe, the gravitational pull on the matter over the Roche lobe from the companion star is going to be stronger than the pull from its own weight. Thus, mass will transfer from the originally heavier star to the lighter one. This process, named the Roche lobe overflow, can either result in the matter being absorbed by the other star or forming an accretion disk, with the latter typically occurring when the accretor is a compact object (Burns Boyle, 1984). It has been theorized that many of these binary systems could not have survived the supernova explosion, in which the primary NS or BH formed, without extensive mass transfer (Tauris & van den Heuvel, 2006b). Due to the probability of the orbit remaining bound after the supernova strongly depends on the mass loss from the system during the explosion. Other factors like the asymmetry of the supernova explosion contribute as well.

3 Modelling and Implementations of theory

The galpy python package is employed for constructing the galactic potential and performing orbital integration. Created by Jon Bovy (Bovy, 2015), it serves as a python package for galactic dynamics. The package provides methods for constructing a galactic gravitational potential from predefined analytic expressions with adjustable free parameters. Additionally, it offers numerous functions for evaluating and integrating crucial quantities regarding these gravitational potentials. It includes a methodology for integrating orbits over time and saving complete orbital information. Furthermore, galpy is affiliated with the astropy package, allowing it to support methods defined there.

3.1 galpy

The galpy package uses a cylindrical galactocentric coordinate system. To convert from a coordinate system centered on the Sun to one centered on the center of the galaxy, one must know the distance of the Sun to the galactic center, as well as its vertical distance above it. Given that the exact location of the Solar System has no impact on investigating the behaviour of binary systems in the galaxy, galpy adopts a slightly different system. When utilizing analytical potentials, the center of the galaxy is defined to be at spherical polar radius $r = 0$. Consequently, by using it as the origin of a cylindrical polar coordinate system, a vertical distance coordinate z and a cylindrical polar radius coordinate R in the $z = 0$ plane can be defined. For quantities, galpy uses its internal units, which are scales defined for each type of quantity. The radius is defined in multiples of $R_0 = 8$ kpc, and the velocity of $v_0 = 220$ kms^{-1}. These values roughly resemble the properties of our solar system within the Milky Way.

3.2 Galactic gravitational potential

As outlined in Section 1, my primary focus, given the current observational frontier, was to model the Milky Way galaxy. In order to accurately model the gravitational potentials while emphasising computational efficiency, I opted for analytical expressions. Each expression presented in Section 2.1 is multiplied by an overall scale, denoted as f_b for the bulk, f_d for the disk and f_h for the dark matter halo. This overall scale is defined in such a way that a scale of one is equivalent of that component having the mass of the Milky Way, as defined in Table 1. The free parameters of the expressions were fine-tuned following Bovy & Rix (2013), and the potentials were scaled by the component masses to resemble the experimentally determined distribution in the Milky Way. The parameters and the overall scale factors are presented in Table 1.

Table 1: The fine tuned parameters for the potentials used. The parameters are defined in the current and Section 2.1.

Parameters	Values
f_d	0.60
f_b	0.05
f_h	0.35
M_d (10^{10} $M_\odot$)	6.8
M_b (10^{10} $M_\odot$)	0.5
$M(r \leq 60$ kpc) (10^{11} $M_\odot$)	4.1
a (kpc)	3.00
b (kpc)	0.28
r_c (kpc)	1.9
R_s (kpc)	16
α	-1.8

The three components were combined with their overall scale to form the combined gravitational potential. The rotation curve for this potential is illustrated in Figure 1.

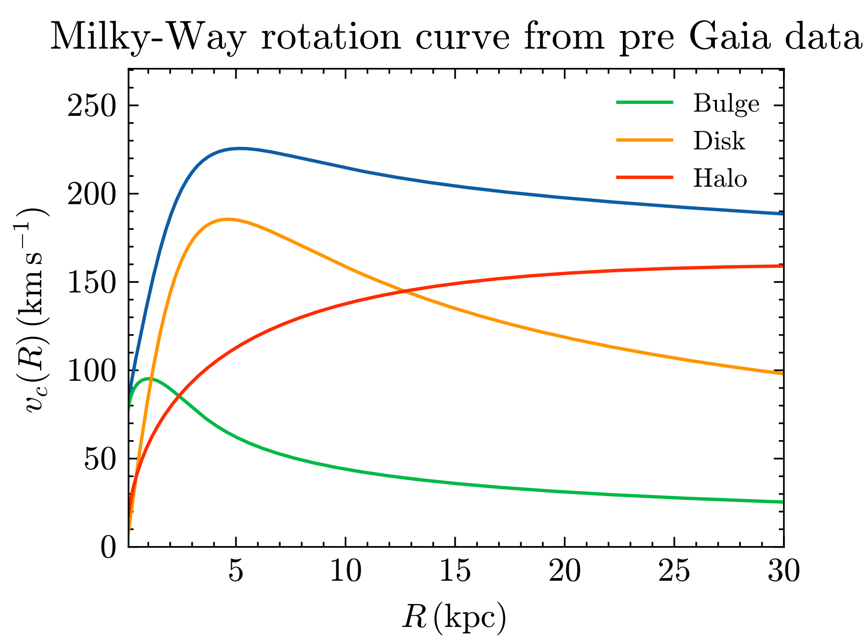

Figure 1: The plot of the rotation curve for the Milky Way galaxy constructed in `galpy` by the Author. The blue curve is the overall rotation curve from the combined potentials.

3.3 Isotropic kick

This project's primary focus revolves around binary systems involving at least one pulsar. Since pulsars are rapidly spinning neutron stars that produce radio emissions from their magnetic poles, these emissions make their observation particularly advantageous. Moreover, my interest lies in system have undergone at least one supernova explosion, thus receiving a natal kick. To understand how systems evolve when subjected to a range of natal kicks, it was essential to determine the direction and ranges of magnitude of these kicks.

Despite lack of significant evidence supporting a preferred direction for natal kicks (Bray & Eldridge, 2018), there are some momentum conservation arguments related to mass ejection (Janka, 2017), but there is no widely accepted standard. Hence, for the sake of generality, I opted to apply kicks in an overall isotropic fashion. The same number of direction vectors was created as the number of systems in the realization of the simulation. To generate random but isotropic vectors, the uniform random distribution was utilized. This involved draining a random number uniformly from the interval [-1,1] for each direction axis of the vector, followed by normalizing the vector to unity. The representations of these vectors can be seen in Figure 2.

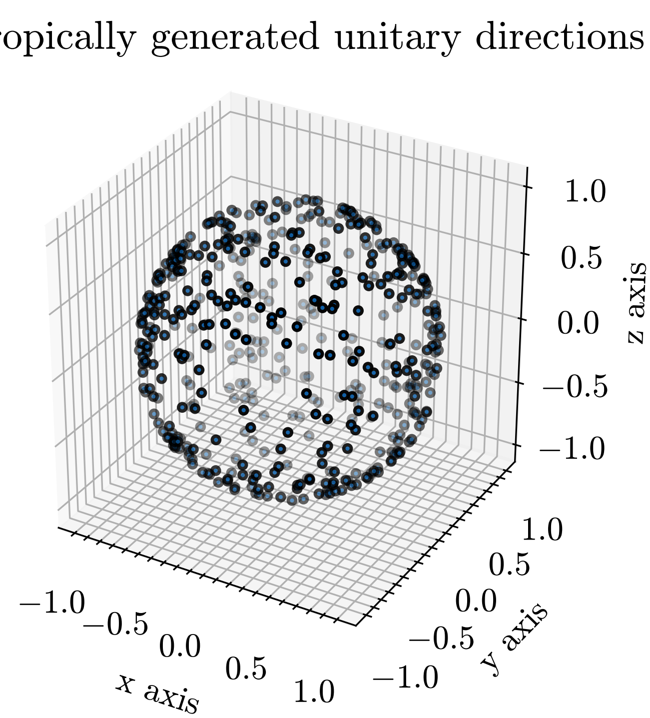

Figure 2: 3D plot representing the isotropicly generated unitary vectors. The three Cartesian axes are used for representation, direction of each vector is denoted by the location and orientation of the circle corresponding to it on the surface of a unit radius sphere.

3.4 Initial binary distribution

When simulating the evolution of binary stars I need to consider an initial distribution of them inside the galaxy. Given the uncertainties surrounding the early matter distributions of galaxies and the lack of comprehensive understanding of the associated physics, I adopted a novel approach. I assumed that the Milky Way galaxy has been static over time, based on its unusually quite history of encounters with other galaxies. The estimated time of the last merger galaxy is approximately 10 Gyr (Yang et al., 2023), resulting in the dark matter content of our galaxy being is less than expected from a spiral galaxy of this size (Hammer et al., 2007). Therefore, my assumption is to consider the current baryonic matter distribution of the Milky Way as unchanged throughout the past. Thus, the initial baryonic matter distribution of the Milky Way is assumed to be the current distribution.

Given that my galactic potential is composed from three components, with one corresponding to the dark matter halo and the remaining two corresponding to the baryonic matter constituting the bulge and the disk, a second assumption is made. Specifically, it is assumed that the stellar mass of the galaxy is equal to the combined mass of its bulge and disk. Studies suggest that this simplification provides a reasonable order of magnitude estimate (Hammer et al., 2007). It is implied that the distribution of stellar mass is the same as the mass distribution of the combined bulge and disk. Give the assumption of a static galaxy, the initial stellar mass distribution must be equivalent to the current stellar mass distribution. This, than can be used as a probability density function (PDF) for the initial distribution of the binary systems inside the Milky Way.

To construct the PDF, a description of the galaxy's mass in terms of the spherical polar radius from its center is required. The analytical expressions of the potentials can be integrated to determine the total mass enclosed up to a given radius. This integration is carried out by using Gauss's law for gravity in the following manner.

$$\oiint_A \mathbf{g} \cdot \mathrm{d}\mathbf{S} = -4\pi G M \tag{4}$$

Where, $\mathbf{g}$ is the gravitational field, $\mathrm{d}\mathbf{S}$ is the infinitesimal vector surface area on the closed surface A through which the integration is done. To include the gravitational potential in this, I can make use of the gravitational field equation, written below.

$$\mathbf{g} = -\nabla\Phi \tag{5}$$

When combining the two equations above, I end up with the desired equation that relates the integral of the gradient of the gravitational potential to the mass enclosed.

$$\oiint_A (\nabla\Phi) \cdot \mathrm{d}\mathbf{S} = 4\pi G M \tag{6}$$

Here, to calculate the mass enclosed up to a radius, we define the surface to be a sphere up to that radius and integrate the gradient of the potential on that surface. From this an enclosed mass versus radius plot was obtained that can be seen in Figure 3.

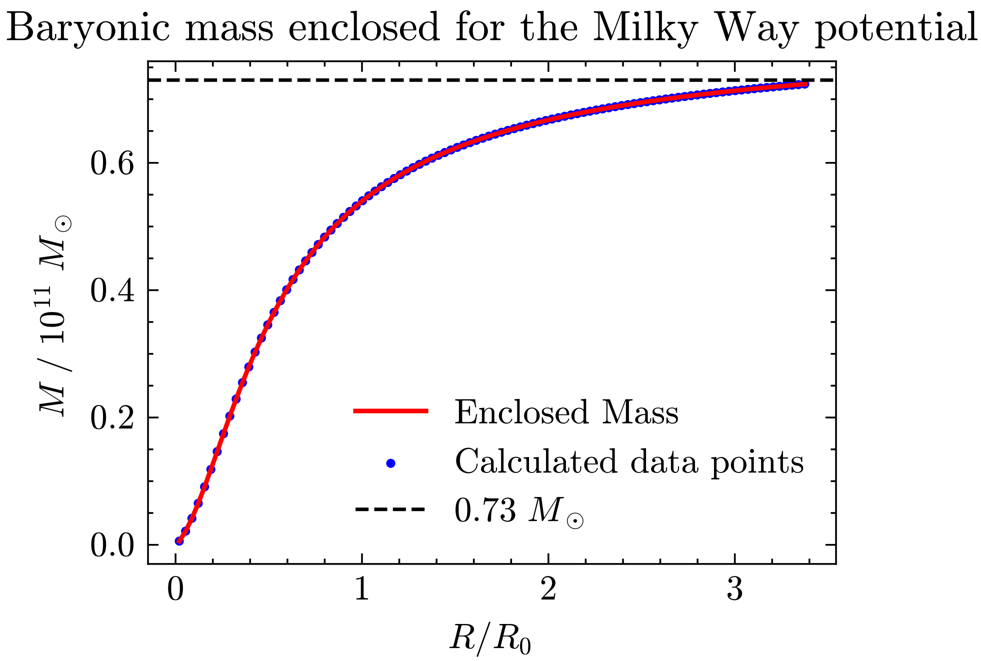

Figure 3: The enclosed baryonic mass plot for the Milky Way potential. On the horizontal axis R_0 is the `galpy` natural unit of distance, 8 kpc. The black dotted line is the total mass obtained from the analytic expression as can be seen in Table 1. I took the radius of the Milky Way to be the radius at which 99% of the mass from the analytical expressions of the disk and the bulge at 60 kpc is enclosed. Hence, the radius of my Milky Way galaxy is 27 kpc.

As mentioned in the description of Figure 3, the radius of my Milky Way galaxy is 27 kpc. This differs from the older conventional value, but recent models based on new measurements were inconsistent with the older, smaller radius and estimated a bigger radius (Xu et al., 2015).
From the enclosed mass plot a histogram of masses was created by binning the mass in the following way. The range of radii was divided into an arbitrary number of equally spaced intervals, the number was chosen to be a hundred. For each of these intervals, the mass of the galaxy in the spherical shell with thickness equal to the interval was calculated from the enclosed mass data points. The radius of the mid point was taken and the mass at each interval was denoted. A function was fitted to these data points and than it was normalized, such that the integral over the interval of 0 to the galactic radius (27 kpc) is equal to unity. This normalized function is than used as the PDF for the initial distribution of binaries, this can bee seen in Figure 4.

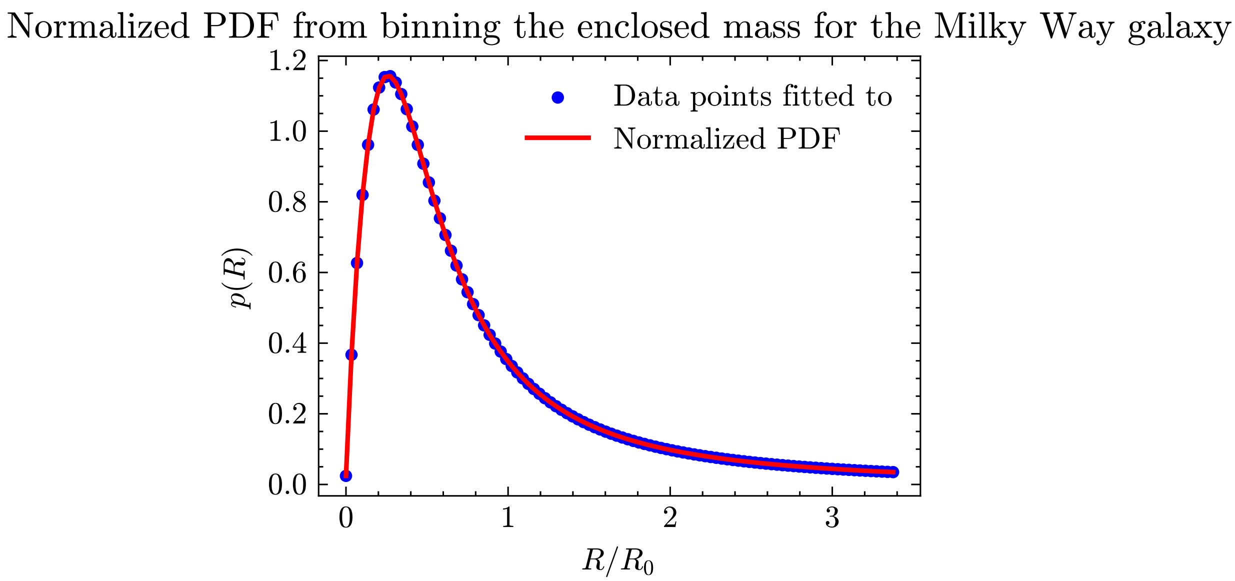

Figure 4: The PDF for the initial distribution of binaries. On the horizontal axis the radius can be seen in natural units, while on the vertical axis the probability density is displayed.

3.5 Generating the binary systems and Orbital integration

The binary systems were generated using the normalized PDF introduced above, using non-uniform random variate sampling techniques. The fit to the distribution was taken as a continuous distribution, rather than using a discrete one with the data points. From the distribution an arbitrary number of binary systems were generated at different radii according to the PDF.

Initially, it was assumed that all binary systems reside in the $z = 0$ plane and that they are revolving on a stable circular orbit. These simplifications were made, because the velocity needed to change from a stable circular orbit to on that has a period in the z coordinate is minute compared to the velocities received during a natal kick. The velocity for a circular orbit was calculated and added to each binary system.

Since the potential used is axisymmetric around the z axis, the initial azimuth angle of the location of the binary is completely arbitrary, hence they were all placed at the same azimuth of $\phi = 0$. Initially, the binaries were evolved for a full circular orbit for the sake of visualisation. Furthermore, a natal kick was applied to the binaries in their initial location. This does not put a bias onto the side of the galaxy at which seeding of the binaries took place, since a rotation of the entire setup by an arbitrary degree is equivalent to putting the binaries at the rotated position.

Lastly, the binary systems were evolved up to Hubble time after the natal kick and the details of their orbits was recorded during the process. The variables recorded during each time steps are the following: the cylindrical polar coordinates centered on the galactic center: radius (R), vertical distance above the plane (z) and the azimuth angle (ϕ) in the $z = 0$

plane, each with the associated velocity (v_R, v_z and v_T).

4 Structure and Results

I created a simulation of the Milky Way galaxy and a method to integrate the time evolution and track the orbits of the binary systems seeded into the galaxy, using the `galpy` python package (Bovy, 2015).

4.1 Code structure

The computer program follows a similar structure to that outlined in Section 3. Except the orbital integration part, almost all main elements of the code are done linearly due to the reliance of information from the previous part. The flow chart representing the process of the simulation can be seen in Figure 5. As the flow chart clearly shows, one of the main goals during the early stages of developing the computer program was to create the PDF from which the binary systems can be generated using random variate sampling. This is highlighted as the blue process on the diagram. It creates a barrier between two distinct components of the program, differing both in structure and logic.

The first part is mainly computational modelling, where I make or use existing mathematical models to describe the investigated physics phenomenon. The programming here is mainly focused on executing the different mathematical calculations necessary to build up the model. All of the tasks accomplished in this part are non-repetitive. In the second section of the code the focus is shifted towards simulation, the integration of the binary orbits is a computationally heavy task, that is repeated for each binary system. The optimization of this part of the code is crucial in decreasing the execution time.

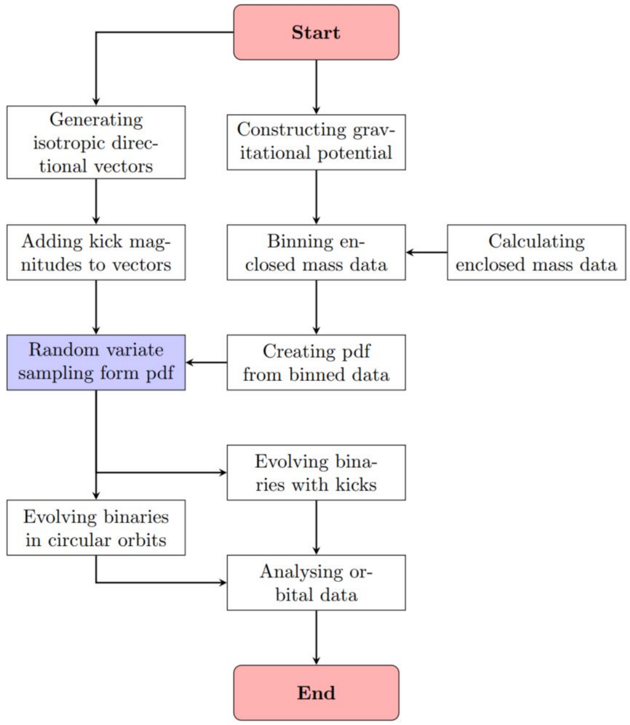

Figure 5: The flow chart of the simulation, in red the star and end parts are visible, between them the workflow of the program can be seen. One of the main sub-goals of the project, where much of the previous work is combined is highlighted in blue.

4.2 Computational Techniques and Features

The program was written using a declaration approach at first, mainly due to the Authors unfamiliarity with the `galpy` python package. This eased the process of determining the use of inbuilt functions and the effective ways of using the package. At later stages a functional approach was adopted to ease the process of running the code and create a more see-through environment for optimization.

For fitting to the binned mass data points to create a PDF, the `UnivariateSpline()` function was used from the `scipy.interpolate` package. This function returns a spline, which is a piecewise polynomial, which is continuous everywhere, hence the resulting spline can be evaluated anywhere like a regular function. However, since the spline is constructed from polynomials on different intervals, it does not behave like a single one-to-one function would, outside of the intervals of interest. Thus, one needs to be careful when defining the interval over which the spline will be used.

The spline was fitted from 0 kpc to 27 kpc, so this is the interval needed to normalize over to create the the desired PDF. The normalization was done in the process of creating the sampler, for which the `rv_continuous()` base class from the `scipy.stats` package was used. The `rv_continuous()` is a continuous random variable base class meant for subclassing.

When creating a new random variable distribution at least the `_pdf()` method has to be overwritten, the function in this method needs to be properly normalized as well. I normalized the distribution during subclassing, such that the integral of the spline form 0 kpc to 27 kpc is equated to unity. In Figure 6 a snippet of the subclassing code is shown.

```
#subclass created to use random variate sampling
class binary_distribution(rv_continuous):
    def __init__(self, name='binary', a=0., b=GALACTIC_RADIUS/8., res=BIN_NUMBER):
        # a and b give the limits of the galaxy

        super(binary_distribution, self).__init__(name=name,a=a,b=b)

        # defining the binary distribution as the spline fit
        self.binary_dist = spline

        self.x = np.linspace(a,b,res)
        self.y = self.binary_dist(self.x)

        # calculating the integral in the range of the galaxy
        self.norm = spline.integral(self.a,self.b)

    # overwriting the _pdf() method, while normalising the distribution
    def _pdf(self, x):
        return self.binary_dist(x)/self.norm

# instance of the class created
spline_dist = binary_distribution()

# binary stars generated using the above written random variate sampler
star = spline_dist.rvs(size=SAMPLE_SIZE)
```

Figure 6: Snippet of my subclassing code for sampling from a custom PDF. Here the `spline` has been constructed in a function written beforehand and passed down to the current one. The overwriting of the `_pdf()` with the normalized distribution can be seen. This produces the PDF shown in Figure 1. Below the class, `spline_dist` is the initialization of the subclass, while `star` is a list of binary stars sampled form the distribution.

When doing the orbital integration part, the `Obrit.integrate()` function is used from the `galpy.orbit` library, but first an `Orbit` object has to be initialized. It can be initialized using a list of 6D phase space coordinates, which are as follows: $[R, v_R, v_T, z, v_z, \phi]$, the coordinates are the same as outlined in Section 3.5. It is possible to initialize multiple orbits at once using a list of the lists of 6D phase space coordinates. Internal units are used for giving the quantities of the phase space, since `galpy` works the most robustly when using its internal units (Bovy, 2015). To integrate orbits a list of times has to be defined, which is then used as epochs for evaluating the orbital variables. Here, a list of equally spaced values between the star and end point are generated and `astropy.units` is used for specifying the times in dimensional units. Initializing orbits together comes with an advantage during integration, an entire collection of orbits can be integrated rapidly if the time interval is the same for all. This makes use of parallel computing and decreases the execution time of the integration process significantly. Although, integrating the kicked orbits for a Hubble time makes extensive use of this, the initial evolution of the binary systems needs to be integrated separately, since the initial evolution times vary for each.

4.3 Results

To classify the calculated orbit for a binary system the phase space variables can be used. Initially, a plot of selected two-dimensional phase spaces was created for visualization purposes, as depicted in Figure 6. The constraints imposed by the axisymmetric nature of the potentials are clearly visible in the phase space diagrams. The orbit of the binary system illustrated is a bound orbit, evident from the rosette shape in the $x - y$ phase space on the top-left figure. The red curve represents the original circular orbit, denoted by the red dots on the bottom panels. The right frame at the bottom is referred to as surfaces of section and was first introduced by Poincaré. The red dot on this frame is the consequence of the shell orbit, where the star's trajectory is confined to a two-dimensional surface. The shell orbit is the limit of the green orbits on the bottom-left frame. If the curve on the bottom-right panel were a single curve folding in on itself, it would suggest the existence of a third integral of motion. The other two arise from the conservation of the energy of the orbit after the kick and the conservation of a component of orbital angular momentum.

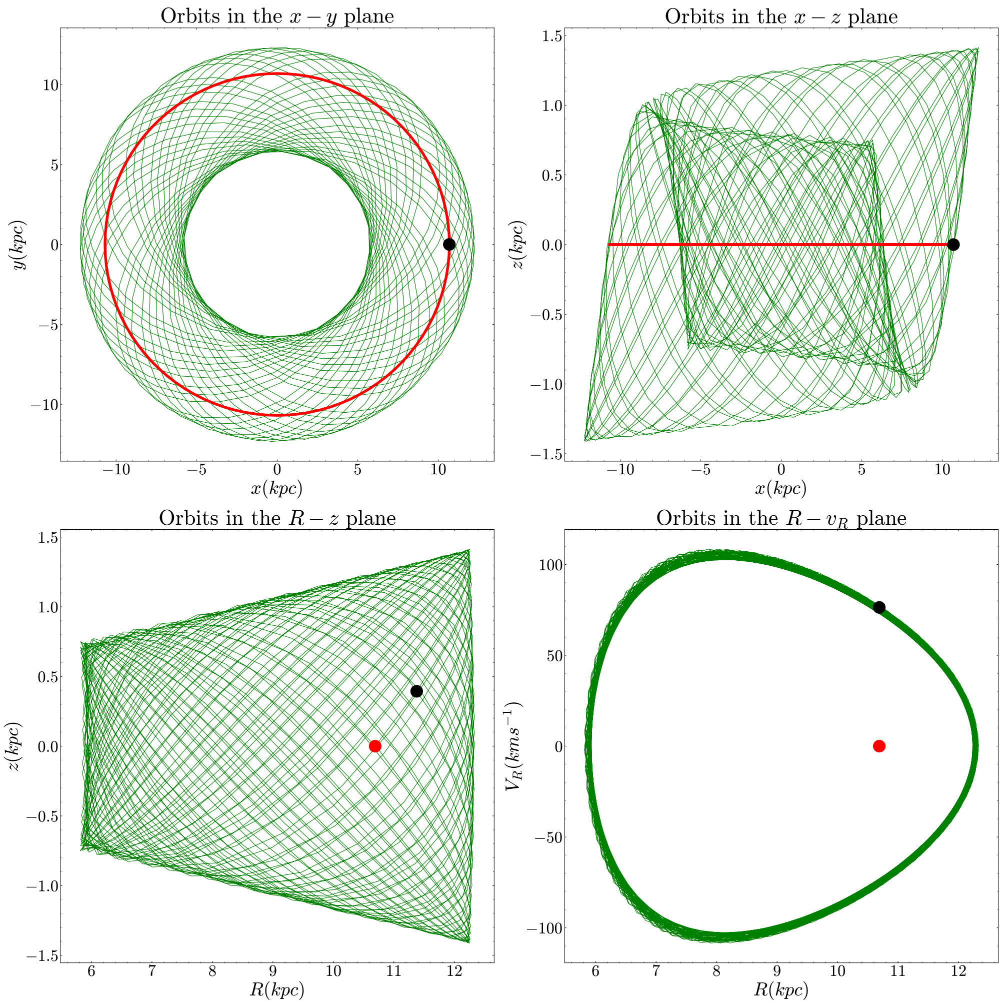

Figure 7: Here a bound orbit can bee seen, which has the classic guitar pick shape in the $R-v_R$ plane. The red curve and point denotes the phase space location of the initial circular orbit and for the top two figures black dot represents the point at which the kick was applied, while on the bottom two plots the black dot is placed at the values of the first time step after the kick. The magnitude of the kick velocity used was 100 kms^{-1}.

5 Discussion and Further improvements

In the future of the project there are a handful of things listed for improvements and further investigation. Analysis of the current techniques and possible improvements of them are discussed in Section 5.1 and suggestions and plans for further work are listed in Section 5.2.

5.1 Overview of approximations and current modelling techniques

During the modelling a number of approximations were made, here I review them and if applicable give suggestions of possible improvements that could be made. Firstly, the galac-

tic potential was modelled by axysimetric expressions. These describe the overall shape of the Milky Way relatively well, while the computational power required to work with them is minute compared to N-body simulations. Those can model a galaxy much more precisely, then analytical expressions, thus the characteristic dynamics of spiral galaxies are present in them. For my investigations of overall trends this could prove very useful, but there are plenty of more improvements to be made beforehand.
Secondly, for modelling the isotropic kicks a bimodal Maxwellian distribution could be adopted to incorporate the recent models fitted to observations. Furthermore, introducing some kind of random bias to the kicks would provide a ground for some interesting investigations.
Lastly, the assumptions made for creating the initial binary population were simplistic compared to the current models of star formation rate in early galaxies and the research into the initial mass function of stellar population. Moreover, it would be interesting to compare my initial radial distribution of binaries obtained to those in other works (Breivik et al., 2020; Mandhai et al., 2022). Also, incorporating the fact that heavy mass binary systems are likely to be born in globular clusters could lead to a more realistic scenario.

5.2 Plan of further work

In Section 4.2 I have mentioned the upgrade of moving from a declaration approach to a functional after establishing a basic proficiency with the `galpy` package. The next improvement here to make is to rewrite the program into an object-oriented approach to ease up the construction of different simulations. This is one of the most urgent improvements scheduled for the future, since there are plans for a lot of further simulations.
The way of analysing results and classifying orbits is subpar at the moment, once a larger set of data will be obtained this needs to be improved and reworked significantly. In order to do this, metrics are needed to be created, that encompass the orbital information of a binary and group them based on certain properties. One way to do this is to implement a machine learning model trained to group different binaries together based on their orbital properties.

One of the improvements developed currently is to extend the simulation to include binary neutron stars, where both of the components of the compact binary systems undergo a supernova explosion and a natal kick. The binaries that remain intact throughout this process are particularly intriguing, and their evolution involves a merger at some point after the formation of both NS. These systems can be initially generated from stellar evolution codes, like COSMIC (Breivik et al., 2020) and BPASS (Eldridge et al., 2017) and then seeded into my model and evolved. The primary objective for the continuation of the project is to prime the locations of these systems in the sky, based on our simulations. My primary focus is going to be determining these predictions for spider pulsars (Fruchter et al., 1988). Additionally, I plan to explore the contamination of binaries in the Milky Way from satellite galaxies and vice versa. This will involve the investigation of the interactions between satellite galaxies and the Milky Way.

6 Summary

As stated in the beginning an analytical description for the potentials was constructed using the NFW, MN and SPL with exponential decay potential profiles. Binary systems were generated using random variate sampling from the PDF obtained from the current distribution of baryonic matter in the Milky Way. These systems were evolved initially on circular trajectories until a natal kick velocity was added to each. These kicked systems then evolved for a Hubble time and then it was investigated if they remain bound and how does their orbital properties change. Subsequently, possible ideas for further research and improvements to the simulation were discussed in detail. The analysis of double neutron star systems is outline for future reference.

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